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A comprehensive global review of country-level food composition databases using a food science frame of reference

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ABSTRACT

Background: Country-level food composition databases should aim to be national resources that contain the nutrient values of the food produced and eaten within a country. The basis of the databases is food science, including the chemical analysis of the nutrients, and the processing and cooking methods used.
Scope and approach: This review undertook a rigorous, systematic search to obtain country-level food composition databases available online from across the world. It found 107 databases and reviewed them for 80 categories including types of foods, number and groups of nutrients, cooking and processing methods and the origins of their data.
Key findings and conclusions: Most food composition databases contain traditional foods (104) and raw plant (100) and animal (96) foods, but only a few contain newer food types such as meat analogues (20). The most included cooking and processing methods were all traditional or long-established practices of dried (97), fermented (93), oil (91) and ground foods (89), while newer and more technologically advanced methods were far less represented. 68 databases included analysed data from the chemical analysis of foods, whilst 46 include imputed or calculated values which are not primary data from food analysis. Using food science as a frame of reference to review food composition databases shows challenges include terminology, and keeping up with new technologies in food analysis and food processing. Food composition databases should be scientifically sound, nationally relevant and user-friendly so they are valuable national resources about a country's food supply.

1. Introduction

Food composition databases should aim to be important national resources that document the nutritional composition of the foods commonly consumed in each country (Bell et al., 2011). They play an important role in individual and population health and can be utilised in many ways. Food composition databases contribute to Sustainable Development Goal (SDG) 2 – Zero hunger, and are especially important in their contribution to Target 2.2 to end all forms of malnutrition (United Nations, 2015). They strengthen national food security by providing information about the nutrition in a nation's food supply for foods that are socio-culturally appropriate.

Food composition databases are used by many different professions, but their primary use is by nutritionists and dieticians to give dietary advice to their patients, for planning and assessing diets, to perform dietary intake studies, to estimate intake of specific nutrients and for clinical research (Cruz et al., 2007; McNutt et al., 2008; Institute of Nutrition, 2007; Pennington, 2008). Researchers use them for studies involving diets and nutrient intake, for dietary surveys, for epidemiology studies and to understand impacts that changes to food and consumption have on populations (Ahuja et al., 2013; Lupiañez-Barbero et al., 2018). They are used by the food industry in the development and optimisation of food products, to support health claims, for food labelling and to calculate reference intakes (Almaamari et al., 2024; Pennington, 2008; Roodenburg & Leenen, 2007). Governments and policy makers use them as resources for making dietary guidelines and recommendations, to make public health interventions, to help with feeding programs, for national food security, to plan appropriate production, for ensuring population diets have nutritional value and for food trade (Ahuja et al., 2013; Church, 2005; Institute of Nutrition, 2007; Ocké et al., 2021; Office for Resources, 2015; Schönfeldt et al., 2018). They can help with identifying edible plants, recognising under-utilised or over-looked foods, selecting foods to improve nutrition by using a wider range of foods, finding scientific names of foods

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enabling access to more information sources, and with selecting new crops to grow (French & Food Plants International, 2025). They can be used to increase diversity in edible plants used in diets which can potentially help to improve nutrition (Cantwell-Jones et al., 2022).

Food science studies the chemical, physical and biological makeup of foods and the properties and behaviours of food including composition, processes, safety and quality (IFST, 2026). It is food science, and specifically food chemistry, which analyses food samples and provides the nutrient data that is presented in food composition databases (Fyfe et al., 2022; Md Noh et al., 2020). Food chemists create new and more advanced methods for the analysis of the nutrients within foods (Striegel et al., 2018). Food science includes food processing which studies how foods change and is applied to develop new food products and to improve existing products (IFST, 2026). Food scientists are involved in food processing in the food industry and in the creation of new methods for food processing.

Despite their many uses and their importance as national resources, food composition databases have not been well studied from a food science perspective. There are other studies which have been performed to understand food composition databases across the world. The study by Brinkley et al. (2025) looked at a number of databases for their general data attributes, some basic food and nutritional data, the food groups they contain, and for their findability, accessibility, interoperability and reusability as data sources. However, their study included very limited content from a food science perspective. In 2023 FAO/INFOODS released an evaluation framework that can be used to assess the quality of food composition databases (Charrondiere et al., 2023). This resource can be used to analyse the quality of a specific database, by surveying its database information, components present, units used, food groups present, and some data information. But despite its breadth it does not include food processing methods or the sources of the data. Its focus is on the standardisation of databases for compilers and users, but not the quality of the data from a food science or food industry perspective.

Food composition databases are made at the country, regional and global levels, depending on their purpose and who is developing them. They are most often made at the country-level as resources of nutrient data of the foods the country produces and eats and to provide data that is relevant to that specific country. Country-level databases are able to include traditional and unique foods that represent their dietary and culinary traditions (Opatha et al., 2025) in addition to the conventional processed foods that are more widely eaten.

This review takes food science as its frame of reference for understanding and analysing food composition databases. Using a food science lens is justified because the content in food composition databases is based on food science. The food descriptions in many databases contain information about the preparation or form of the food which includes the cooking and processing methods used to prepare them. Food chemical analysis is used to obtain the values for nutrient data of each food. Despite this, food composition databases are seen as the domain of nutritionists and database compilers rather than food scientists which has resulted in a research gap around food composition databases from a food science perspective. This review aims to assist in overcoming that research gap, by looking globally at food composition databases at the country-level and analysing them from a food science perspective for the types of foods, nutrients and processing methods they contain, and the sources of their data.

2. Methodology

2.1. Search for food composition databases

Food composition databases in this study were obtained from online sources. Some were found using online lists of databases including the International Network of Food Data Systems (INFOODS) directory (FAO, 2024) and through membership of the European Food Information Resource (EuroFIR) (EuroFIR AISBL Secretariat, 2025). Because this study was analysing food composition databases at the country-level, a rigorous online search was performed using an English language list of all the countries in the world. Databases for each country were searched for online using all three of the following search terms with the name of each country inserted in the 'country name' section:

- 'Country name' food composition database
- 'Country name' food nutrition database
- 'Country name' nutrition composition

Webpages found in the search were screened to see if they contained a food composition database. Food composition databases were either downloaded and then analysed for the review, or were analysed online, depending on their format. A full list of databases included in this study is in Table S1 in the supplementary material. To keep this study relevant for countries this analysis included both government and private databases at the country-level and excluded databases that were at the regional or global level or were comprised of foodomics data. The search and analysis took place between April 2024 and January 2026.

2.2. Database review

To comprehensively review them, each database was analysed for a total of 80 different categories including 12 types of foods, 10 types of nutrients (components), 47 types of cooking and processing methods and six types of data sources. Indicator variables were used - if the type of food/nutrient/cooking or processing method were present it was recorded as a 'yes', and if it wasn't it was recorded as a 'no'. The total number of foods and the total number of nutrients each database contained was recorded. The data origins section analysed the types of data sources of the data based on the information available, including information in documentation, the tables or web pages. The categories the databases were analysed for and explanations of each one are provided in Table 1. Google translate (https://translate.google.com) was used to translate into English those databases that were in other languages and which did not contain an English translation. The results were entered into a Microsoft Excel spreadsheet (Version 2406, Microsoft 365, Microsoft Corporation, USA). Statistics and graphs were performed using Excel. Maps to visualise data globally were made using Excel which uses information from Bing, the Australian Bureau of Statistics, GeoNames, Microsoft, Navinfo, Open Places, OpenStreetMap, Overture Maps Fundation, TomTom and Zenrin to create the maps.

There were several limitations with this analysis. The analysis was performed in English and so the names and descriptions of foods in non-English language databases relied on the translation. Finding the total number of nutrients was sometimes difficult in online databases where nutrients could only be looked at by individual foods. Therefore, it is possible, although unlikely, that some databases do contain more nutrients then shown in the results.

3. Results

3.1. About the databases

The search for databases found over 140 databases globally across the country, region and global levels, but excluded those at region and global levels to ensure relevance at the national level. The initial search found 111 databases at the country-level, and of these 107 were able to be analysed. Of the four that were excluded, two could not be accessed, one could not be translated, and one could not be opened. Table 2 and Fig. 1 provide more information about the databases. Fig. 1a shows the countries that had between 1 and 4 databases in this study. Table 2 shows 85 databases had names of foods in languages other than English (LOTE), and 64 contained scientific names of plant and/or animal foods. The mean number of foods was 5581 (Table 2), however, this number

Table 1
List of categories that databases were analysed for with explanatory notes.

Name	Explanatory note
Database information	
Total foods	Total number of foods in the database
LOTE	Names of foods are in a language other than English (LOTE), includes those with English names alongside the LOTE names
Scientific names	Scientific names are given for raw plant or animal foods
Food types	
Raw plants	Food sourced from edible plants that have not been cooked or processed
Raw animals	Food sourced from animals that have not been cooked or processed
Beverages	Beverages other than juice and milk
Branded	Brand names are present in the description of foods
Meals	Meals or composite foods
Traditional foods	Meals, food products or foods including indigenous foods with a long history of production and consumption within that country's food culture
Ultra-processed	Foods that would be classified as ultra-processed
Fortified	Foods that have been fortified or enriched including those with added vitamins or minerals
Instant	Foods designed for rapid preparation that contain the word 'instant' in the name or description
Meat analogues	Meat analogues, meat alternatives and plant-based meat
Dairy analogues	Dairy analogues, dairy alternatives, plant-based milks and dairy substitutes
Sugar substitutes	Sugar substitutes, sugar alternatives, non-sugar sweeteners
Nutrient types	
Total nutrients	Total number of nutrients/components in the database
Proximates	Includes values for 1 or more proximates
Minerals	Includes values for 1 or more minerals
Trace Elements	Includes values for 1 or more trace elements
Vitamin C	Includes values for Vitamin C
Folate	Includes values for folate
Vitamins	Includes values for 1 or more vitamins
Sugars	Includes values for individual sugars, not just total sugars
Organic acids	Includes values for individual organic acid, not just total organic acids
Fatty acids	Includes values for individual fatty acids, not just totals
Amino acids	Includes values for individual amino acids, not just total
All nutrient types	The database contains levels of some nutrients of all the groups above
Cooking and Processing Methods	
Ground	Ground or flour
Dried	Dried
Dried and ground	Dried and ground or powdered
Broken/flakes	Undergone processing so they are broken, flaked or grit
Rolled	Rolled as a processing method, i.e. rolled oats but not rolled pork
Hulled	Grains that have the bran removed described as hulled, decorticated, polished or bran
Canned	Canned or tinned
Pickled	Pickled
Frozen	Frozen or deep-frozen
Concentrated	Concentrated foods including evaporated
Puree	Puree, mash or paste
Oil	Oil
Juice	Juice
Fermented	Fermented foods including cheese
Boiled	Boiled for any length of time, e.g. boiled, simmered, par-boiled, blanched
Fried	Fried or pan-fried but not deep-fried
Deep-fried	Deep-fried
Baked	Baked or par-baked including bread
Toasted	Toasted
Roasted	Roasted
Cooked	Described as cooked but without the cooking method
Grilled	Grilled or broiled
Barbecued	Cooked by barbecue, not barbecue flavour or barbecue sauce
Smoked	Smoked
Casseroled	Casseroled
Stewed	Stewed
Poached	Poached
Steamed	Steamed
Cured	Cured foods including salami and bacon and foods salted for preservation
Meat fat	Meat fat or blubber, but not oils or butter
Trimmed	Meat with fat removed i.e. trimmed or lean
Extracted	Extracts of foods including starch, tapioca and vanilla, but not beverages, juice or oil
Pasteurised	Pasteurised
Skimmed	Milk described as skimmed, trim or reduced fat
UHT	Ultra-high temperature (UHT) processing
Curd	Curd
Microwaved	Microwaved
Pressure-Cooked	Pressure or pressure-cooked
Pressure separation	Mechanical separation of meat using pressure
HPP	High pressure processing (HPP)
Cold-pressed	Cold pressure
Carbonated	Carbonated beverages including soda, cola and soft drink
Extruded	Described as extruded, not just described as pasta
Puffed/popped	Puffed or popped
Protein isolate	Protein isolate
Hydrogenated	Hydrogenated
Hydrolysed	Hydrolysed
Data Origins	
Analysed	Data is from the chemical analysis of samples of the described foods
Other FCD's	Data is from other food composition databases (FCD)
Literature	Data is from journal articles and other literature
Imputed or calculated	Data has been imputed or calculated from other foods, not analysed for the food described
Product information	Data is from food labels, product information or manufacturers
Not specified	The source of the data has not been specified for some or all of the data

Table 2
Information about the databases.

Database information	Number of databases
Number of databases analysed	107
Databases with language other than English (LOTE)	85
Databases containing scientific names	64
Average number of foods in databases (mean ± SD)	5581 ± 45085
Average number of foods in databases (mean ± SD) excluding USDA	1225 ± 1625

was greatly skewed by the presence of the USDA *Food Data Central* database which contained 467 290 food items (USDA, 2023), far more than any other database. Excluding the USDA, the average number of foods was 1225, ranging from 50 in the *Senegal Food Composition Table* (Yoo et al., 2019), to 10 013 in *Tabula: Spanish Food and Beverage Composition Tables* (CEU University Institute of Food and Society, 2025). The range is seen in Fig. 1b which shows variations in the green colour for this range, but with the much higher USDA database represented in black. The average number of nutrients in the databases was 56, however, there was a relatively large standard deviation of 51. Fig. 1c shows the range in numbers of nutrients from 5 in the *Latvian National Food Composition Data Base* (Food Center of Food and Veterinary Service, 2009), up to 290 in the *Australian Food Composition Database* (FSANZ, 2022). Six of the databases contained more than 150 nutrients.

3.2. Types of food

Fig. 2 shows the number of databases containing each type of food. Most databases contained raw plant (100) and raw animal foods (96), although some contained only processed food products. Of the 107 databases, 104 contained traditional foods of some kind, either traditional meals, Indigenous foods or traditional foods or food products. Of the

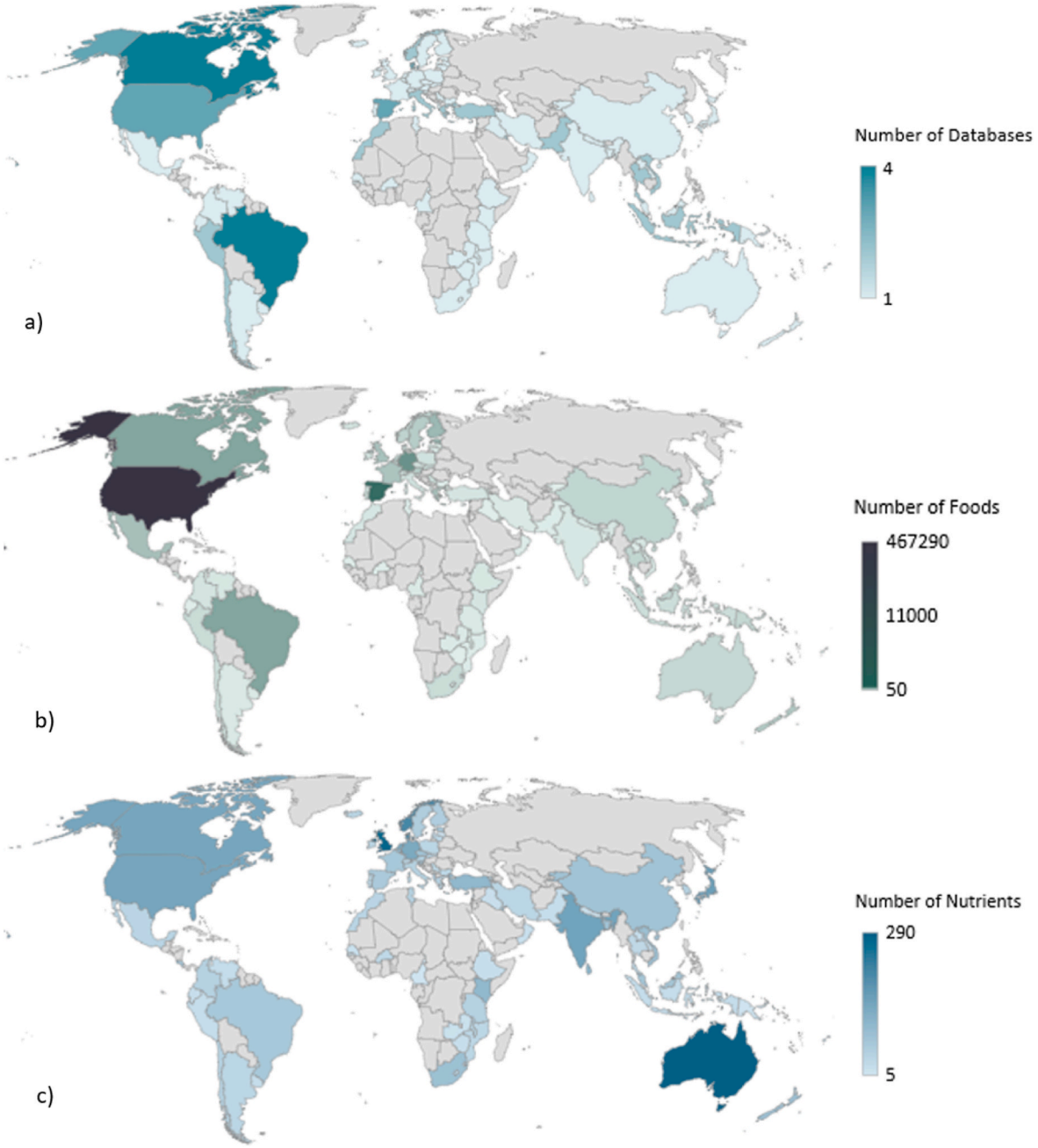


Fig. 1. Country-level comparison of food composition databases showing a) the number of databases per country, b) the number of foods in the databases per country with the highest number represented, and c) the number of nutrients in the databases per country with the highest number represented. Map lines delineate study areas and do not necessarily depict accepted national boundaries.

alternative food products, 20 contained meat analogues such as plant-based meats, 32 contained sugar substitutes, and 54 contained dairy analogues, with many databases including soy milk.

3.3. Nutrients included

Fig. 3 shows that of the 107 databases, most contained proximates (104), minerals (102) and vitamins (98). Less than half of them contained individual fatty acids (42), individual sugars (28), individual amino acids (27) and individual organic acids (12). Eight out of the 107 databases contained values of nutrients for all these nutrient groups. As well as analysing for nutrient groups, this study also analysed for two individual nutrients, vitamin C and folate, which are both important nutrients required for health. Of the 107 databases, 95 contained levels for Vitamin C and 73 included folate.

3.4. Cooking and processing methods included

Fig. 4 presents the number of databases containing each cooking and processing method in their food descriptions, with the methods ordered from highest to lowest number of databases. This shows the most included cooking and processing methods are all long-standing methods that have been used for centuries as ways of preserving or cooking foods. The most included methods are drying (97), fermenting (93), extracting oil (91) and grinding (89). Following these are the long-established cooking methods of roasting (86), baking (85) and frying (85), and the extraction of juice (85). The graph shows that newer and more technologically advanced cooking and processing methods are less

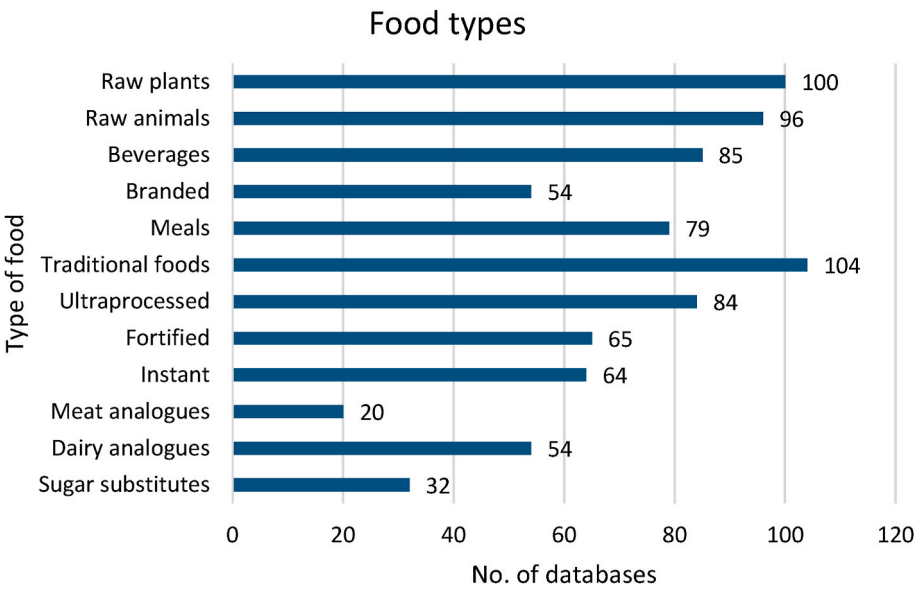


Fig. 2. The different types of foods in the databases and the number of databases containing each type.

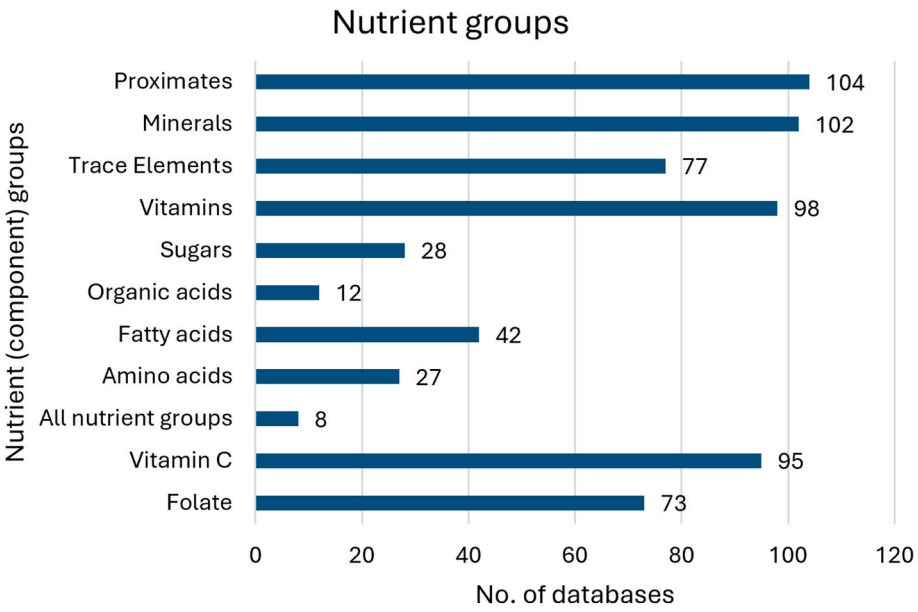


Fig. 3. The number of databases containing each nutrient (component) group and vitamin C and folate.

included in food composition databases. 28 databases contained composition data for protein isolates, 24 for microwaved foods and one database included high pressure processing as a food description.

3.5. Data origins

Fig. 5 shows the analysis of the origins of the data within the 107 databases. Some databases contained a single type of data origin, however, most had a mixture of types and sources of data. 68 databases contained at least some primary analysed data and 14 said all of their data was from food analysed for that database, whilst 39 databases did not contain any primary data. 63 databases said some or all their data came from other databases, with many using data from the USDA database, and most using data from multiple databases. 52 used data that came from literature, usually from peer-reviewed journal articles. 20 databases had data from food labels or from product information provided by the manufacturer. 46 databases contained data that was imputed or calculated. The data origins from 18 databases were either not specified by the database or were not able to be found or translated by the researchers in this study. Some of these 18 contained data from other origins as well as data described as not specified. Overall, the results show there is a lack of primary data within the databases.

4. Discussion

4.1. About the databases

The rigorous search for databases in this study found 107 at the country-level that could be analysed. This was more than the 101 evaluated by Brinkley et al. (2025) which included country, regional and global databases as well as foodomics databases, single food databases and supplement databases. It was also more than the number of databases globally presented in an overview by FAO who found over 100 at both the country and regional level (Grande et al., 2025).

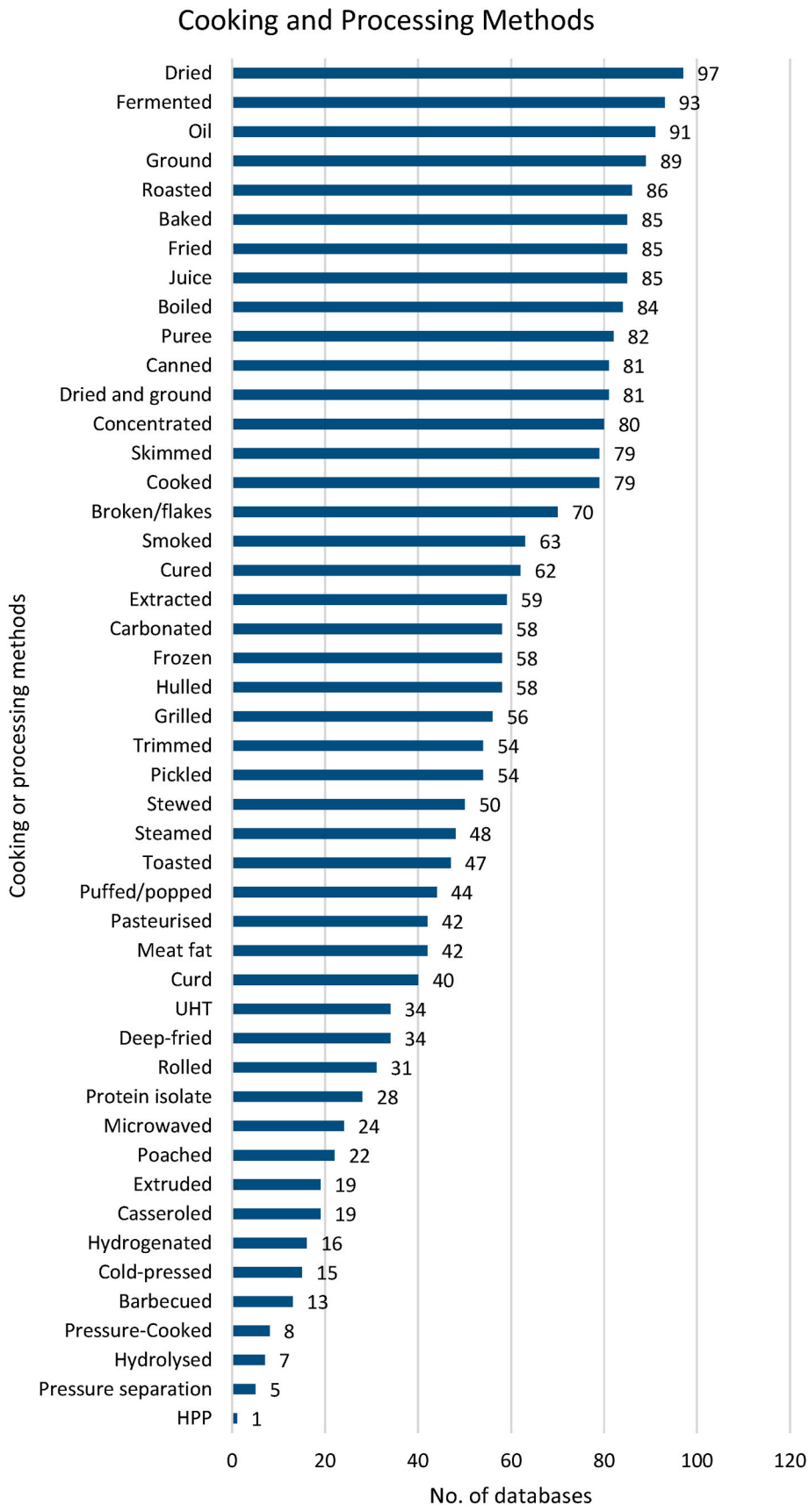


Fig. 4. Number of databases at the country-level containing different types of food cooking and processing methods in their food descriptions.

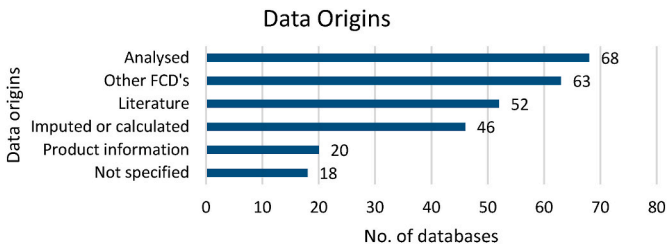


Fig. 5. Number of databases at the country-level containing different types of origins of their data. Many databases contain data from multiple different types of origins so may be included in multiple groups.

Despite only analysing databases at the country-level, this study found there was great variation between them, even for those sourced from the same country. For example, Canada has the *Canadian Nutrient File* which is a larger database with 143 nutrients and 5690 foods that was designed for use by health professionals and the food industry (Health Canada, 2015). Canada also has a smaller database, the *Nutrient Value of Some Common Foods*, which contains 19 nutrients per food and 1100 foods, which was produced to be more accessible for the Canadian people (Health Canada, 2008).

4.2. Types of food

Country-level food composition databases include the currently commonly consumed foods for each nation. This analysis shows that most food composition databases contain values of nutrients for raw plant and animal foods, beverages, meals, traditional foods and ultra-processed foods (Fig. 1). This paper assessed the types of foods included in food composition databases but did not assess the inclusion of food groups as this has been previously undertaken (Brinkley et al., 2025). Traditional foods were in 104 of the 107 databases at the country-level, showing that traditional foods play a major role in diets globally. The inclusion of traditional and indigenous foods within food composition databases shows their importance in diet diversity, food security, nutrition and culture. Food composition databases act as reservoirs of food knowledge, so incorporating traditional and indigenous foods into them can directly affect the foods continued use and cultural preservation and enable their health benefits to be promoted to prevent non-communicable diseases (Costa et al., 2010; Opatha et al., 2025).

Ultra-processed foods are usually not labelled as ultra-processed in food composition databases, but of the 107 databases in this study there were 84 containing foods the authors identified as being ultra-processed. In Fig. 1 most of the foods that contain the word ‘instant’, or are fortified, are meat and dairy analogues or have sugar substitutes would also be considered as ultra-processed. Ultra-processed foods go through multiple stages of processing that modify them, fractionate them, add additives and then rebuild them (Monteiro et al., 2019). The high levels of processing disintegrate food matrices, change protein and fibre networks, cause losses of vitamins and minerals, and change chemical composition and protein structures (Fardet, 2018; Greenfield & Southgate, 2003; Office for Resources, 2015). The extensive processing used to make ultra-processed meat products can lead to protein cross-linking and aggregation and the formation of potentially harmful substances (Yang et al., 2025). All these processes and changes result in food products with different quantities of nutrients to their original ingredients.

Meat and dairy analogues are often portrayed as being healthy alternatives, but their composition is considerably different to meat and dairy and different between analogues (Marchese et al., 2024), which can potentially have health and nutrition effects for consumers. Plant-based meat analogues are often ultra-processed foods made to mimic the nutritional profile of a food but which have achieved this through significant changes through their processing, and with multiple additives (Bohrer, 2019; Martín-Miguélez et al., 2024). These changes are often not represented through food composition data. A study in Australia found that two-thirds of dairy alternatives and one third of meat alternatives sold in grocery stores were not present in the *Australian Food Composition Database* (Marchese et al., 2024). One database attempts to address this, the *Establishment of Food Composition Database for Local Processed Foods in the State of Kuwait* database has classified the foods by the NOVA classification to understand the prevalence of ultra-processed foods (Al-Faili & Al-Mutairi, 2022). The NOVA classification system classifies foods as unprocessed and minimally processed foods (group 1), processed culinary ingredients (group 2), processed foods (group 3) and ultra-processed foods (group 4) (Monteiro et al., 2019). This database added an extra column to their composition tables for the NOVA classification for each food so users of the database can clearly see which foods are ultra-processed, and to enable researchers and policymakers to establish preventive actions to decrease prevalence and severity of non-communicable diseases in population health (Al-Faili & Al-Mutairi, 2022).

4.3. Nutrients included

The results in Fig. 1c and 3 show that databases at the country-level provide nutrient (component) values for different nutrients and classes of nutrients. Some databases present the values of the same nutrients for all foods. Others have appendixes for some nutrients that are only reported for some of the foods within the database. A number of databases, especially online ones, present values of different nutrients for different foods, often with different numbers of nutrients for each food. The *Nutrient Value of Some Common Foods* by Health Canada (2008) chose to provide 19 nutrient values for all foods but they varied what those nutrients were depending on the type of food. Their decision was based on picking the nutrients that were relevant to specific foods e.g. they chose to present cholesterol levels for meats but not fruits and vegetables, and alcohol and caffeine values for beverages. The *Australian Food Composition Database* had the largest number of nutrients in its spreadsheet (290), but most foods only had values for some nutrients (FSANZ, 2022). Likewise, the *Food Composition Tables for Kingdom of Bahrain* gives some types of nutrients for some of the foods but not for all of them (Musaiger, 2011). Another difference was explained by the purpose of the FDA *Total Diet Study* which aims to monitor the level of nutrients and contaminants in current foods being eaten in the US. The database produced from this study only contains minerals, trace elements and heavy metals (FDA, 2022). Despite this, the FDA studies have been used by nutrition professionals as a source of food nutrient data (Pennington, 2008). Differences between nutrients included in databases, and in the methods used for obtaining nutrient values, makes comparisons between databases and data difficult to perform (Uusitalo et al., 2011).

One major difference between food science and food composition database compilation is the terminology used for nutrients. Food scientists call the nutrients in foods ‘nutrients’ and in food composition databases they are called ‘components’. In the food composition database guidelines by INFOODS, they use component identifiers and tagnames as the terminology for identifying nutrients (Klensin et al., 1989). While tagnames and identifiers are used by database compilers, their inclusion should not impede the databases usability. An observation from this study is that databases that use tagnames don't always provide the definitions for them. Providing standardised nutrient names increases usability of the databases for users from many backgrounds and would keep them consistent with the results of food composition analysis presented in food science studies. Food composition databases should be useable and accessible for people across many health, government and professional disciplines, as well as for the general public. Consistent and understandable terminology and language, with nutrient names written clearly and accurately will make the data accessible and able to be utilised by all users.

Over recent years the knowledge about food composition has

increased dramatically through the use of new analysers and technology. Foodomics is the study of food and nutrition using omics technologies (Cifuentes, 2009), and has greatly increased the understanding of the composition of food due to the greater sensitivity of these omics analysis. Metabolomics can be used to study foods to analyse the metabolites, or small chemical compounds, that make up foods (Herrero et al., 2012). Metabolomics analysis has shown that foods are composed of thousands of small compounds, with all food metabolites estimated to be 500 000 (Rychlik & Schmitt-Kopplin, 2020). The *FOODB* database is an online food composition database that uses metabolomics data and it contains 70 926 compounds (Wishart Node TMIC). This is much higher than the 290 nutrients (components) in the *Australian Food Composition Database* (FSANZ, 2022), the highest number at country-level, and it is much higher than the 451 in the FAO INFOODS Biodiversity database at the global-level (Charrondière et al., 2013). Proteomics analyses the proteins contained in foods and can give structural information about these food components. The *Bovine Milk Protein Database* (BoMiProt) contains 10 642 proteins that are found in bovine milk (Das et al., 2022; Maity et al., 2020). The Periodic Table of Food Initiative aims to add another database online with standardised foodomics data and automated data processing tools (Jarvis et al., 2024).

As foodomics analysis increases and our understanding of all the metabolites in foods continues to grow, these newer methods will continue to give much greater insight into foods, and their nutrition and health benefits. These new technologies also provide challenges for food composition databases in presenting and communicating large quantities of food composition data. Food composition databases should be compiled and formatted in such a way that they are user-friendly with good ease of use, intuitiveness and learnability (Clancy et al., 2015). As foodomics increases our understanding of food composition the formatting of the databases to make this increased knowledge accessible to users will need to be considered. Compilers of food composition databases will need to consider what nutrients are included and how databases are formatted to provide the best resources for users and for health and nutrition outcomes.

4.4. Cooking and processing methods included

This study analysed databases for the types of cooking and processing methods they included as food science includes food processing as well as food composition. The processing methods used on food are important inclusions in food descriptions in food composition databases as processing changes foods nutritional composition. By presenting the foods in their raw and in a variety of processed forms the databases show how cooking and processing affects the nutrient composition of that specific food. The processing description and the nutrient values provide more accurate information for users for making food choices which can result in better nutrition and health outcomes. The FAO/INFOODS Evaluation framework has three key pieces of information they say should be included in the description of foods: the state the food is in with regards to preparation, the part or source of the food and how the food has been cooked or processed (Charrondiere et al., 2023). Including composition data for foods that have been processed or cooked using a variety of methods is important as food nutrient changes occur during cooking and processing due to nutrient retention and losses during cooking and processing (Fardet, 2018; Raita et al., 2025; USDA Nutrient Data Laboratory, 2007), and because the matrix of the food containing the nutrients also changes (Aguilera, 2025; Fardet, 2024). These changes to food composition include loss of heat-sensitive vitamins, degradation of nutrients, modifications in protein and fibre structures, increases and decreases in moisture content, the generation of new compounds, modifications to food structure and enzyme modification of nutrients (Institute of Food Technologists, 2026). Other processes, such as fermentation, can increase the quantity of nutrients in a food (Fardet, 2018; Raita et al., 2025). Processing can also affect the digestibility of nutrients within foods, some by enhancing it, and other processes, often associated with ultra-processed foods, by decreasing it (Yang et al., 2025).

A key finding in this analysis is seen in Fig. 4 which shows the number of databases that include each cooking and processing method. The long-standing and more traditional methods of cooking and preserving foods were present in most of the databases but newer and more advanced methods and technologies are included in far fewer databases. Only one database, the *USDA Food Data Central* (USDA, 2023), included all 47 cooking and processing methods in Fig. 4, and it was the only database that included high-pressure processing (HPP). Food processing is done to foods to increase the food safety, to preserve food and extend its shelf-life, to ensure food is available all year round, to make food more convenient, to provide cost-effective products and to assist with dietary patterns (Institute of Food Technologists, 2026). To provide these outputs, food technology is continually advancing, and new processing methods are being developed and incorporated into the food industry. A recent foresight exercise by FAO on new food sources and production systems identified 44 innovations in food that are likely to become available and prevalent in the food supply (Mukherjee et al., 2025), and these would require composition analysis and new entries into databases. As new methods of food processing are used by the food industry it's important that food composition databases provide information about the effects that these processing methods have on the nutrients of foods in the food supply. This study found that only 24 of the 107 databases included microwaved in the description of foods, yet microwave ovens are now a household appliance across the world and are often used on a daily basis in household food preparation. The foods within a food composition database should represent a country's food supply and the food that is eaten by the people in that country. As new and more advanced food processing technologies are used in the country's food industry, the databases should be updated to include these foods and processing methods. An observation in this analysis was that many of the descriptions used in food composition databases include the state of the food, but not all include the processing or cooking method used to take it to this state. For example, a lot of foods are described as ‘dried’ but most don't include the method used to dry it - dried food can be oven dried, spray dried, freeze dried, sundried, blast oven dried, dried by dry frying, dried in a dehydrator etc. and all of these can affect the composition of the nutrients. ‘Dried’ tells the user that moisture has been removed but does not indicate what nutrient changes have occurred due to the drying method. Including the name of the processing method with the form of the food may be useful for some database users.

4.5. Data origins

The origins of the data in food composition databases is important for the traceability of the data and for users to know the provenance of the data and, therefore, the quality. The results of the data origins in Fig. 5 shows that the range of origins includes original analysis, literature, other food composition databases, imputation, calculation, food product information and some which were not specified. The origins of the data and how it is generated reflect differences in data quality. In the book *Food composition data: production, management and use,* Greenfield and Southgate (2003) have a preferential order for data sources starting with original analytical values, then imputed values, then calculated values, then borrowed values and lastly presumed values. The German *BLS* database describes 13 different categories of data sources in their database: analysis, recipe calculations, pattern calculations, literature, aggregation, label information, nutrient database, adopted value, rescaling, logical zero, logical assumption, traces and formula calculation (Max Rubner Institute, 2025).

The most accurate and reliable data is the direct method of performing the original scientific analysis of the nutrients of samples for the database (Greenfield & Southgate, 2003), which enables both the nutrient values and the food description to be accurate. The chemical

analysis of food provides reliable composition data and are usually based on standard methods of analysis (AOAC International, 2023). For a national resource, having all data as scientific analysis is ideal as it means the database will be accurate, reliable and a true representation of the nutrient composition of the foods from that country. However, analysis of samples is expensive and requires advanced technical instrumentation and knowledge (Finglas et al., 2017; Lundberg-Hallén & Öhrvik, 2015; Md Noh et al., 2020; Micha et al., 2018), but the instrumentation required is now common in science laboratories which should make it easier to find ways to prioritise food composition analysis for national databases.

In Fig. 5 the category of calculated/imputed data includes data generated from calculations such as recipe, formula and pattern calculations, imputed data, adopted value's, rescaling, logical zero's, logical assumptions and rescaling. The data in these categories are not analysed data from food, they are values based on calculations and assumptions. Imputed data is data that has been taken, or 'borrowed', from analytical values of a similar food, a different form of that food or from calculations; and these numbers are taken and entered into the database as a nutrient value (Greenfield & Southgate, 2003; Ispirova et al., 2019). Imputed data uses the assumption that the results of one food are the same as the results of another. Calculated data are nutrient levels that have been generated using the values of the ingredients or other forms of the food and then calculated using nutrition calculators or nutrient retention factors and/or yield factors (Reinivuo et al., 2009). They have been described as a 'rough estimate' as the yield and retention of nutrients changes dramatically during processing and cooking (Greenfield & Southgate, 2003). Comparison studies between calculations and analysis show there were some significant differences between the calculations and the results from the analysis in proximates, minerals and vitamins (Silva et al., 2021).

Imputation, calculations and associated methods assume that the results of one sample are the same as the result of another sample, and the use of these methods can undermine the value of the data in food composition databases. Data made by calculating from recipes and raw ingredient data can be less accurate as they don't take into account losses of nutrients through processing or interactions between molecules, especially during ultra-processing which has multiple processes (Opatha et al., 2025). Imputation may be the only option for a nutrition practitioner with a patient sitting with them and no data for a food being discussed, however, for a national resource to use imputed data does not value what a food composition database is or how it is used. The analysis by Brinkley et al. (2025) highlighted some of the risks that occurs when secondary data is relied on, especially data inaccuracies, not being reflective of current local food supplies, and not having the information or data that comes with advances in science. Other errors that occur from borrowing data include misunderstanding how the nutrients have been defined, calculation errors, using incorrect conversion factors, carelessness and ignorance (Ispirova et al., 2020; Micha et al., 2018).

The recycling and estimation of data that occurs in food composition databases has previously been called out as a practice that would be considered inacceptable in many other disciplines (Micha et al., 2018), and would also be inacceptable in food science. Incorrect data in food composition databases can lead to incorrect policies, health claims and food labelling, and to inadequate food choices (Delgado et al., 2021). The process of using imputation negates the scientific rigour involved in the analysis of foods and thereby reduces the value of a database. From a scientific point of view if there are no composition results for a food, then the composition of that food is not known. The data in food composition databases should be of sufficient quality for them to be a reliable source of nutrient values. Ideally, all data in food composition databases should be reliable scientific values of nutrients from the analysis of the food sample described. A food composition database that is based on accurate, reliable data that reflects the actual nutrients in a food supply can be used as a national reference by nutrition, science, health and policy workers across the country or region.

5. Conclusion

The purpose of compiling country-level food composition databases should be to provide accurate nutrient values of foods that are commonly consumed in that country in a way that is accessible and understandable for professional and consumer users. Databases should ideally use primary data from the rigorous scientific chemical analysis of foods so they are valuable and relevant national resources. The compilation of food composition databases has challenges as technologies in food scientific analysis and food processing continue to advance. Databases that are updated to stay relevant to the food supply in a country and to the advances in science, technology and processing within the countries food industry and homes will have much wider use and impact. Formatting databases to reflect this will ensure the data remains accessible and available for all users. Food composition databases should be scientifically sound, nationally relevant and user-friendly so they are valuable national resources about a country's current food supply.

Author contributions

Selina Fyfe: Conceptualisation, Methodology, Formal Analysis, Investigation, Writing – Original Draft, Writing – Review and Editing and Visualisation. **Damian Hine:** Conceptualisation, Resources, Writing – Review and Editing, Supervision, Project Administration and Funding Acquisition. **Yasmina Sultanbawa:** Conceptualisation, Resources, Writing – Review and Editing, Supervision, Project Administration and Funding Acquisition.

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Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.tifs.2026.106108.

Data availability

Data will be made available on request.

References

Aguilera, J. M. (2025). Food matrices as delivery units of nutrients in processed foods. *Journal of Food Science, 90*(2), Article e70049.

Ahuja, J. K. C., Moshfegh, A. J., Holden, J. M., & Harris, E. (2013). USDA food and nutrient databases provide the infrastructure for food and nutrition research, policy, and practice. *The Journal of Nutrition, 143*(2), 241S–249S. https://doi.org/10.3945/jn.112.170043

Al-Faili, B., & Al-Mutairi, A. (2022). *Establishment of food composition database for local processed foods in the State of Kuwait*. Kuwait Institute for Scientific Research.

Almaamari, S., Al-Jawaldeh, A., Al Ghammari, I., Al Shammakhi, S., Al Aamri, J., & El Ati, J. (2024). Nutritional data on selected food products consumed in Oman: An update of the food composition table and use for future food consumption surveys. *Foods, 13*(5), 787.

AOAC International. (2023). In G. W. J. Latimer (Ed.), *Official methods of analysis of AOAC international*. Oxford University Press.

Bell, S., Colombani, P. C., Pakkala, H., Christensen, T., Møller, A., & Finglas, P. M. (2011). Food composition data: Identifying new uses, approaching new users. *Journal of Food Composition and Analysis, 24*(4), 727–731. https://doi.org/10.1016/j.jfca.2011.03.001

Bohrer, B. M. (2019). An investigation of the formulation and nutritional composition of modern meat analogue products. *Food Science and Human Wellness, 8*(4), 320–329.
Brinkley, S., Gallo-Franco, J. J., Vázquez-Manjarrez, N., Chaura, J., Quartey, N. K. A., Toulabi, S. B., Odenkirk, M. T., Jermendi, E., Laporte, M.-A., Lutterodt, H. E., Annan, R. A., Barboza, M., Amare, E., Srichamnong, W., Jaramillo-Botero, A., Kennedy, G., Bertoldo, J., Prenni, J. E., Rajasekharan, M., … Ahmed, S. (2025). The state of food composition databases: data attributes and FAIR data harmonization in the era of digital innovation [Original Research]. *Frontiers in Nutrition, 12*. https://doi.org/10.3389/fnut.2025.1552367
Cantwell-Jones, A., Ball, J., Collar, D., Diazgranados, M., Douglas, R., Forest, F., Hawkins, J., Howes, M.-J. R., Ulian, T., Vaitla, B., & Pironon, S. (2022). Global plant diversity as a reservoir of micronutrients for humanity. *Nature Plants, 8*(3), 225–232. https://doi.org/10.1038/s41477-022-01100-6
CEU University Institute of Food and Society. (2025). *Food and beverage composition tables of Spain TABULA*. CEU University Institute of Food and Society. https://ias.ceu.es/tabula-bbdd/presentacion/.
Charrondiere, U. R., Stadlmayr, B., Grande, F., Vincent, A., Oseredczuk, M., Sivakumaran, S., Puwastien, P., Judprasong, K., Haytowitz, D., & Gnagnarella, P. (2023). *FAO/INFOODS evaluation framework to assess the quality of published food composition tables and databases*.
Charrondière, U. R., Stadlmayr, B., Rittenschober, D., Mouille, B., Nilsson, E., Medhammar, E., Olango, T., Eisenwagen, S., Persijn, D., & Ebanks, K. (2013). FAO/INFOODS food composition database for biodiversity. *Food Chemistry, 140*(3), 408–412.
Church, S. (2005). The history of European food composition databases. *European Food Information Resource Consortium (EuroFIR)*.
Cifuentes, A. (2009). Food analysis and Foodomics. *Journal of Chromatography A, 1216*(43), 7109. https://doi.org/10.1016/j.chroma.2009.09.018
Clancy, A. K., Woods, K., McMahon, A., & Probst, Y. (2015). Food composition database format and structure: A user focused approach. *PLoS One, 10*(11), Article e0142137. https://doi.org/10.1371/journal.pone.0142137
Costa, H. S., Vasilopoulou, E., Trichopoulou, A., Finglas, P., & on behalf of the participants of the Euro, F. I. R. T. F. W. P.. (2010). New nutritional data on traditional foods for European food composition databases. *European Journal of Clinical Nutrition, 64*(3), S73–S81. https://doi.org/10.1038/ejcn.2010.215
Cruz, A. J., Cervera, P., & Gascon, M. B. (2007). *Tabla de composición de alimentos*. Spain: Novartis Medical Nutrition.
Das, A., Giri, K., Behera, R. N., Maity, S., & Ambatipudi, K. (2022). BoMiProt 2.0: An update of the bovine milk protein database. *Journal of Proteomics, 267*, Article 104696. https://doi.org/10.1016/j.jprot.2022.104696
Delgado, A., Issaoui, M., Vieira, M. C., Saraiva de Carvalho, I., & Fardet, A. (2021). Food composition databases: Does it matter to human health? *Nutrients, 13*(8), 2816. https://www.mdpi.com/2072-6643/13/8/2816.
EuroFIR AISBL Secretariat. (2025). *European food information resource (EuroFIR)*. EuroFIR AISBL Secretariat. https://www.eurofir.org/.
Fardet, A. (2018). Chapter three - Characterization of the degree of food processing in relation with its health potential and effects. In F. Toldrá (Ed.), *Advances in food and nutrition research, 85* pp. 79–129). Academic Press. https://doi.org/10.1016/bs.afnr.2018.02.002.
Fardet, A. (2024). Ultra-processing should be understood as a holistic issue, from food matrix, to dietary patterns, food scoring, and food systems. *Journal of Food Science, 89*(7), 4563–4573.
Finglas, P., Roe, M., Pinchen, H., & Astley, S. (2017). The contribution of food composition resources to nutrition science methodology. *Nutrition Bulletin, 42*(3), 198–206.
Food and Agriculture Organization of the United Nations. (2024). International network of food data systems (INFOODS) international food composition table/database directory. *Food and Agriculture Organization of the United Nations*. https://www.fao.org/infoods/infoods/tables-and-databases/en/.
Food and Drug Administration (FDA). (2022). Total diet study report: Fiscal years 2018-2020 elements data. *United States: United States Food and Drug Administration*. Retrieved from https://www.fda.gov/food/fda-total-diet-study-tds/fda-total-diet-study-tds-results.
Food Center of Food and Veterinary Service. (2009). *Latvian national food composition data base 2009. Brussels, Belgium: EuroFIR - European food information resource*.
Food Standards Australia New Zealand. (2022). *Australian food composition database*. Australia: Food Standards Australia New Zealand.
French, B. R., & Food Plants International. (2025). *Food plants international: Edible plants of the world*. Food Plants International. Retrieved 24th July from https://foodplantsinternational.com/.
Fyfe, S., Schirra, H. J., Rychlik, M., van Doorn, A., Tinngi, U., Sultanbawa, Y., & Smyth, H. E. (2022). Future flavours from the past: Sensory and nutritional profiles of green plum (*Buchanania obovata*), red bush apple (*Syzygium suborbiculare*) and wild peach (*Terminalia carpentariae*) from East Arnhem Land, Australia. *Future Foods, 5*, Article 100136.
Grande, F., Rittenschober, D., Nanjo, Y., Moltedo, A., & Holmes, B. (2025). A global overview of the state of food composition tables and databases. In *14th international food data conference*. Rome, Italy.
Greenfield, H., & Southgate, D. A. (2003). *Food composition data: Production, management, and use*. Food & Agriculture Org.
Health Canada. (2008). *Nutrient value of some common foods*. Ottawa, Canada: Health Canada.
Health Canada. (2015). *2015 Canadian nutrient file. Brussels, Belgium: EuroFIR*.
Herrero, M., Simó, C., García-Cañas, V., Ibáñez, E., & Cifuentes, A. (2012). Foodomics: MS-based strategies in modern food science and nutrition. *Mass Spectrometry Reviews, 31*(1), 49–69. https://doi.org/10.1002/mas.20335
Institute of Food Science and Technology. (2026). Food science and food technology - Overlapping but different disciplines. *Institute of Food Science and Technology*. Retrieved 13th July from https://www.ifst.org/food-science-food-technology.
Institute of Food Technologists. (2026). Food processing. *Institute of Fod Technologists*. Retrieved 5th February 2026 from https://www.ift.org/policy-and-advocacy/advocacy-toolkits/food-processing.
Institute of Nutrition. (2007). *Vietnamese food composition table - bẢNG thành PHẦN THỰC PHẨM VIỆT NAM*. Vietnam: Medical Publishing House - NHÀ XUẤT BẢN Y HỌC.
Ispirova, G., Eftimov, T., Korošec, P., & Koroušić Seljak, B. (2019). MIGHT: Statistical methodology for missing-data imputation in food composition databases. *Applied Sciences, 9*(19), 4111. https://www.mdpi.com/2076-3417/9/19/4111.
Ispirova, G., Eftimov, T., & Seljak, B. K. (2020). Evaluating missing value imputation methods for food composition databases. *Food and Chemical Toxicology, 141*, Article 111368. https://doi.org/10.1016/j.fct.2020.111368
Jarvis, A., Gallo-Franco, J., Portilla, J., German, B., Debouck, D., Rajasekharan, M., Khoury, C., Herforth, A., Ahmed, S., Tohme, J., Arnaud, E., Golden, C. D., Dawid, C., de Haan, S., DeClerck, F., Feskens, E. J. M., Fogliano, V., Fritz, G., Hald, C., … de la Parra, J. (2024). Periodic Table of Food Initiative for generating biomolecular knowledge of edible biodiversity. *Nature Food, 5*(3), 189–193. https://doi.org/10.1038/s43016-024-00941-y
Klensin, J. C., Feskanich, D., Lin, V., Truswell, A. S., & Southgate, D. A. T. (1989). *Identification of food components for INFOODS data interchange*. The United Nations University. https://archive.unu.edu/unupress/unupbooks/80734e/80734E00.htm.
Lundberg-Hallén, N., & Öhrvik, V. (2015). Key foods in Sweden: Identifying high priority foods for future food composition analysis. *Journal of Food Composition and Analysis, 37*, 51–57. https://doi.org/10.1016/j.jfca.2014.09.008
Lupiañez-Barbero, A., González Blanco, C., & de Leiva Hidalgo, A. (2018). Spanish food composition tables and databases: Need for a gold standard for healthcare professionals. *Endocrinología, Diabetes y Nutrición (English ed.), 65*(6), 361–373. https://doi.org/10.1016/j.endien.2018.05.011
Maity, S., Bhat, A. H., Giri, K., & Ambatipudi, K. (2020). BoMiProt: A database of bovine milk proteins. *Journal of Proteomics, 215*, Article 103648. https://doi.org/10.1016/j.jprot.2020.103648
Marchese, L. E., Hendrie, G. A., McNaughton, S. A., Brooker, P. G., Dickinson, K. M., & Livingstone, K. M. (2024). Comparison of the nutritional composition of supermarket plant-based meat and dairy alternatives with the Australian Food Composition Database. *Journal of Food Composition and Analysis, 129*, Article 106017. https://doi.org/10.1016/j.jfca.2024.106017
Martín-Miguélez, J. M., Martín, I., González-Mohíno, A., Souza Olegario, L., Peromingo, B., & Delgado, J. (2024). Ultra-processed plant-based analogs: Addressing the challenging journey toward health and safety. *Journal of Food Science, 89*(12), 10344–10362.
Max Rubner Institute. (2025). *German federal food code (BLS), version 4.0 - German nutrient database*. Karlsruhe, Germany: Max Rubner Institute.
McNutt, S., Zimmerman, T. P., & Hull, S. G. (2008). Development of food composition databases for food frequency questionnaires (FFQ). *Journal of Food Composition and Analysis, 21*, S20–S26. https://doi.org/10.1016/j.jfca.2007.05.007
Md Noh, M. F., Gunasegavan, R. D.-N., Mustafa Khalid, N., Balasubramaniam, V., Mustar, S., & Abd Rashed, A. (2020). Recent techniques in nutrient analysis for food composition database. *Molecules, 25*(19), 4567. https://www.mdpi.com/1420-3049/25/19/4567.
Micha, R., Coates, J., Leclercq, C., Charrondiere, U. R., & Mozaffarian, D. (2018). Global dietary surveillance: Data gaps and challenges. *Food and Nutrition Bulletin, 39*(2), 175–205. https://doi.org/10.1177/0379572117752986
Monteiro, C. A., Cannon, G., Lawrence, M., Costa Louzada, M.d., & Pereira Machado, P. (2019). *Ultra-processed foods, diet quality, and health using the NOVA classification system, 48*. Rome: FAO.
Monteiro, C. A., Cannon, G., Levy, R. B., Moubarac, J.-C., Louzada, M. L., Rauber, F., Khandpur, N., Cediel, G., Neri, D., & Martinez-Steele, E. (2019). Ultra-processed foods: What they are and how to identify them. *Public Health Nutrition, 22*(5), 936–941.
Mukherjee, K., Trieb, J., Niegowska Conforti, M., Di Martino, M., Fattori, V., & Lipp, M. (2025). *Exploring the future landscape of new food sources and production systems – A foresight exercise*. Rome, Italy: FAO.
Musaiger, A. O. (2011). *Food composition tables for Kingdom of Bahrain*. Bahrain: Arab Center for Nutrition.
Ocké, M. C., Westenbrink, S., van Rossum, C. T. M., Temme, E. H. M., van der Vossen-Wijmenga, W., & Verkaik-Kloosterman, J. (2021). The essential role of food composition databases for public health nutrition – Experiences from the Netherlands. *Journal of Food Composition and Analysis, 101*, Article 103967. https://doi.org/10.1016/j.jfca.2021.103967
Office for Resources. (2015). *Standard tables of food composition in Japan*. Japan. *Ministry of Education, Culture, Sports, Science and Technology (MEXT)*. Japan.
Opatha, J., Dissanayake, L., Herath, B., Jayaweera, K., Wijesinghe, H., Wijesinghe, H., Jabir, S., Sumathipala, A., Rijsdijk, F. V., Zavos, H. M. S., & Jayawardena, R. (2025). Development of a country-specific food composition database for Sri Lanka. *Journal of Food Composition and Analysis, 140*, Article 107230. https://doi.org/10.1016/j.jfca.2025.107230
Pennington, J. A. T. (2008). Applications of food composition data: Data sources and considerations for use. *Journal of Food Composition and Analysis, 21*, S3–S12. https://doi.org/10.1016/j.jfca.2007.02.005
Raita, J., Ahmed, H., Chen, K., Houttu, V., Haikonen, R., Kårlund, A., Kortesniemi, M., Yang, B., Koistinen, V., & Hanhineva, K. (2025). Existing food processing classifications overlook the phytochemical composition of processed plant-based protein-rich foods. *Nature Food*. https://doi.org/10.1038/s43016-025-01148-5

Reinivuo, H., Bell, S., & Ovaskainen, M.-L. (2009). Harmonisation of recipe calculation procedures in European food composition databases. *Journal of Food Composition and Analysis, 22*(5), 410–413. https://doi.org/10.1016/j.jfca.2009.04.003

Roodenburg, A. J. C., & Leenen, R. (2007). How food composition databases can encourage innovation in the food industry. *Trends in Food Science & Technology, 18* (8), 445–449. https://doi.org/10.1016/j.tifs.2006.11.026

Rychlik, M., & Schmitt-Kopplin, P. (2020). Reading from the crystal ball: The laws of moore and kurzweil applied to mass spectrometry in food analysis [Perspective]. *Frontiers in Nutrition, 7*. https://doi.org/10.3389/fnut.2020.00009

Schönfeldt, H. C., Hall, N., & Pretorius, B. (2018). The important role of food composition in policies and programmes for better public health: A South African case study. *Food Chemistry, 238*, 94–100. https://doi.org/10.1016/j.foodchem.2016.12.067

Silva, M., Ribeiro, M., Viegas, O., Martins, Z. E., Faria, M., Casal, S., Pinto, E., Almeida, A., Pinho, O., & Ferreira, I. M. P. L. V. O. (2021). Exploring two food composition databases to estimate nutritional components of whole meals. *Journal of Food Composition and Analysis, 102*, Article 104070. https://doi.org/10.1016/j.jfca.2021.104070

Striegel, L., Chebib, S., Netzel, M. E., & Rychlik, M. (2018). Improved stable isotope dilution assay for dietary folates using LC-MS/MS and its application to strawberries. *Frontiers in chemistry, 6*, 11.

United Nations. (2015). *Transforming our world: The 2030 agenda for sustainable development*. New York, USA: UN General Assembly. Retrieved from https://sdgs.un.org/publications/transforming-our-world-2030-agenda-sustainable-development-17981.

U.S. Department of Agriculture. (2023). *USDA food data central*. USA: U.S. Department of Agriculture.

USDA Nutrient Data Laboratory. (2007). *USDA table of nutrient retention factors: Release 6*. Beltsville, MD, USA: U.S. Department of Agriculture (USDA).

Uusitalo, U., Kronberg-Kippilä, C., Aronsson, C. A., Schakel, S., Schoen, S., Mattisson, I., Reinivuo, H., Silvis, K., Sichert-Hellert, W., Stevens, M., Norris, J. M., & Virtanen, S. M. (2011). Food composition database harmonization for between-country comparisons of nutrient data in the TEDDY study. *Journal of Food Composition and Analysis, 24*(4), 494–505. https://doi.org/10.1016/j.jfca.2011.01.012

Wishart Node TMIC. *The food database (FOODB)*. Canadian institutes of health research, Canada foundation for innovation and the metabolomics innovation centre (TMIC). Retrieved 16th March from www.foodb.ca.

Yang, D., Chen, C., Zhao, D., & Li, C. (2025). Impact of ultra-processed meat products on human health: Review and outlook. *Journal of Food Science, 90*(2), Article e70040.

Yoo, Y. M., Atkin, R., & Pachón, H. (2019). Development of a food composition table to analyze Senegalese food expenditure data. *African Journal of Food, Agriculture, Nutrition and Development, 19*(2).